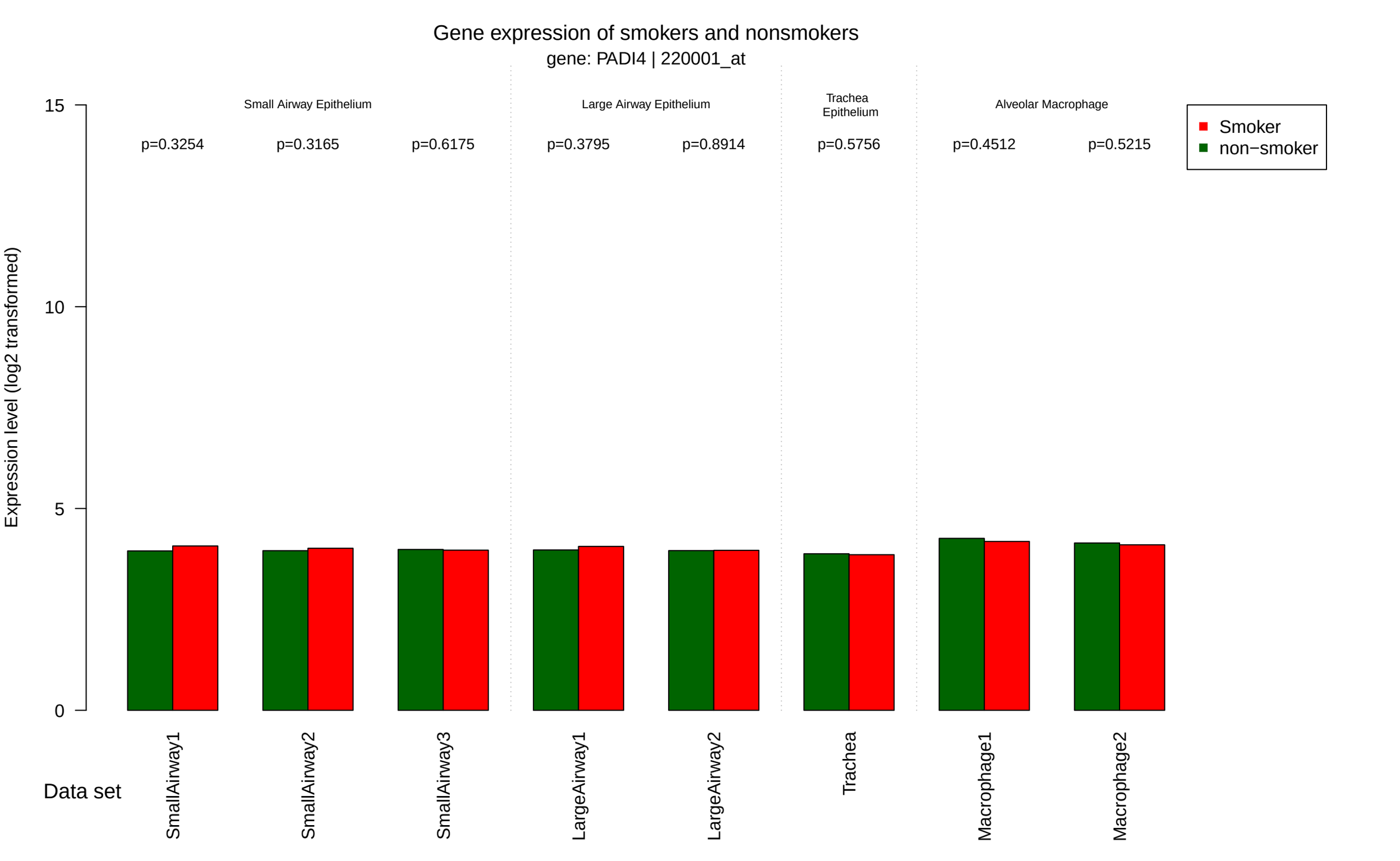
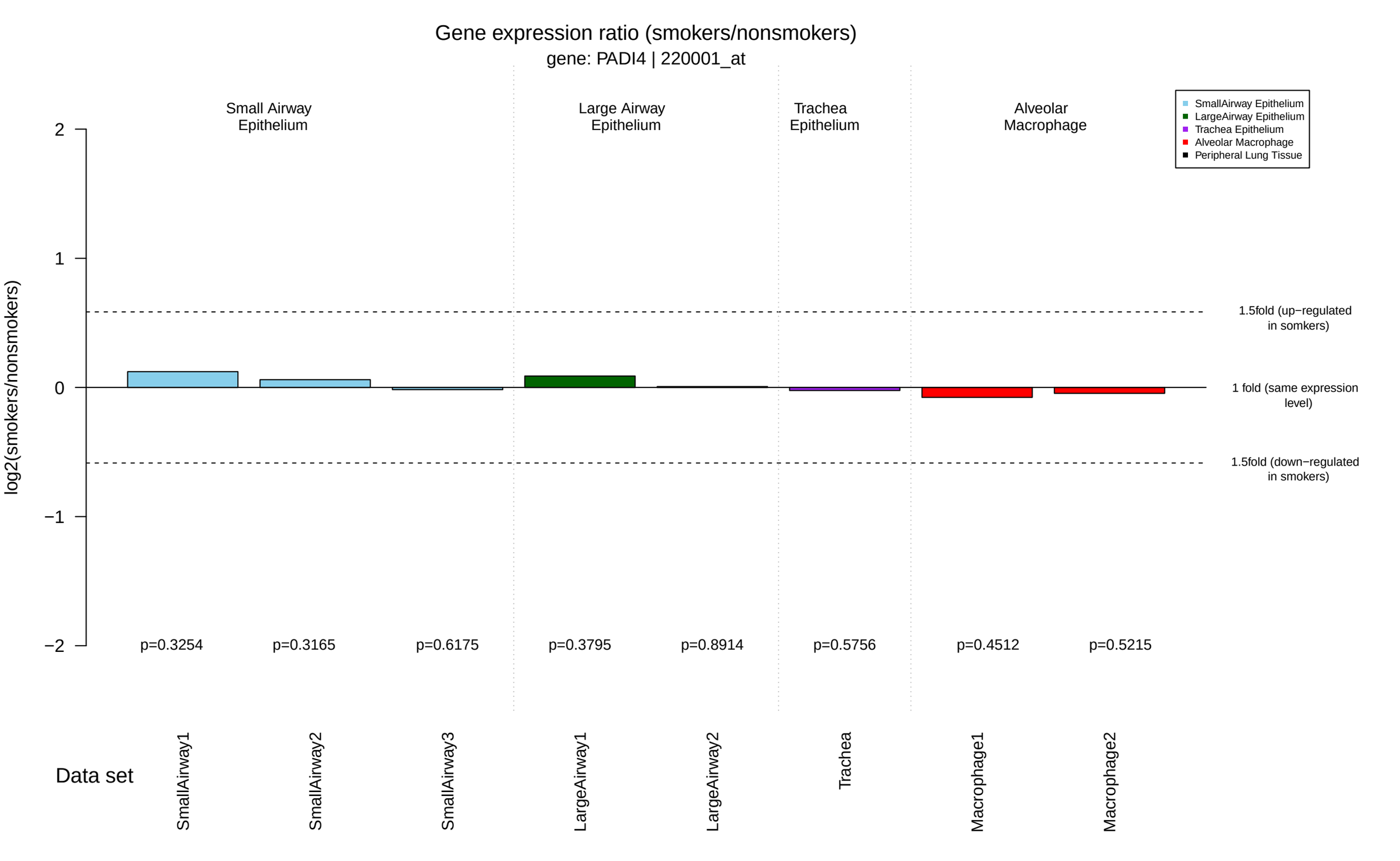
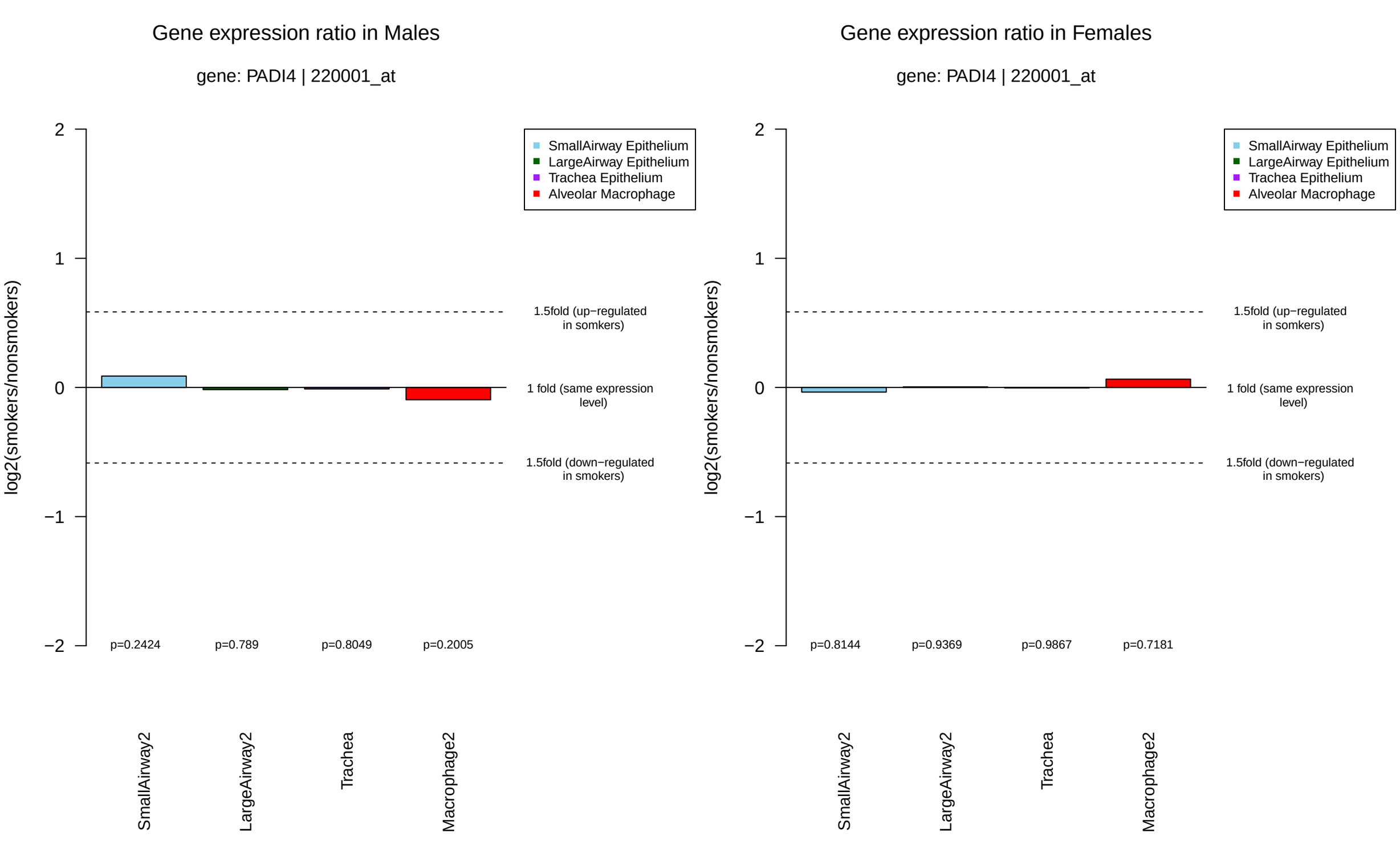
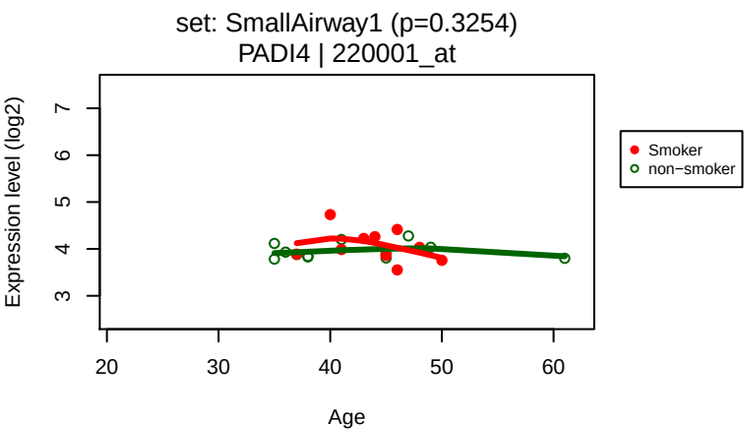
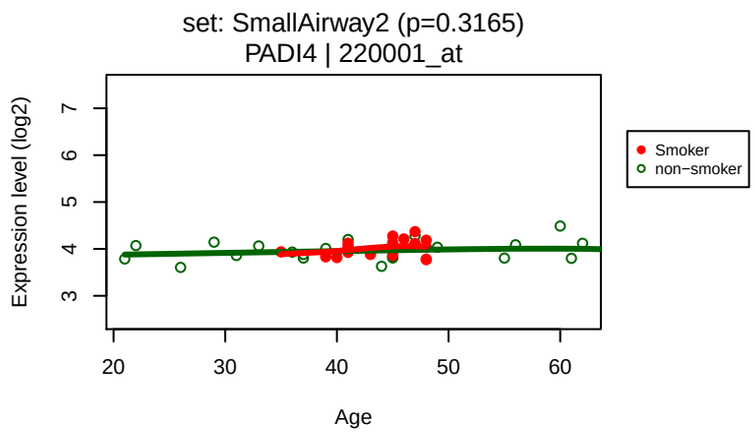
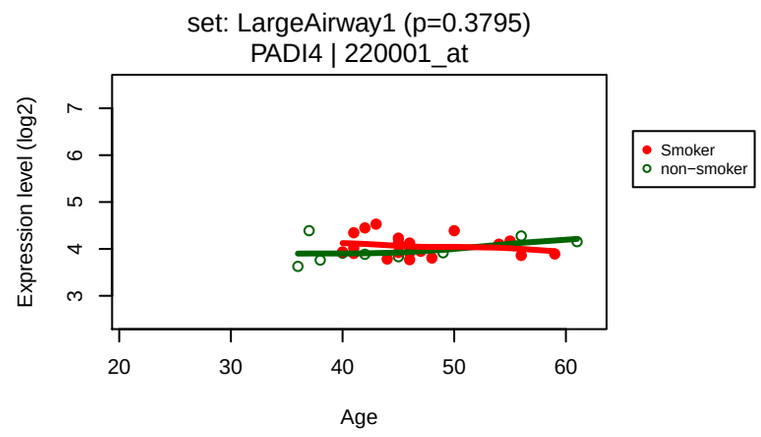

Gene expression of smokers and nonsmokers
gene: PADI4 | 220001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3254
p=0.3165
p=0.6175
p=0.3795
p=0.8914
p=0.5756
p=0.4512
p=0.5215
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

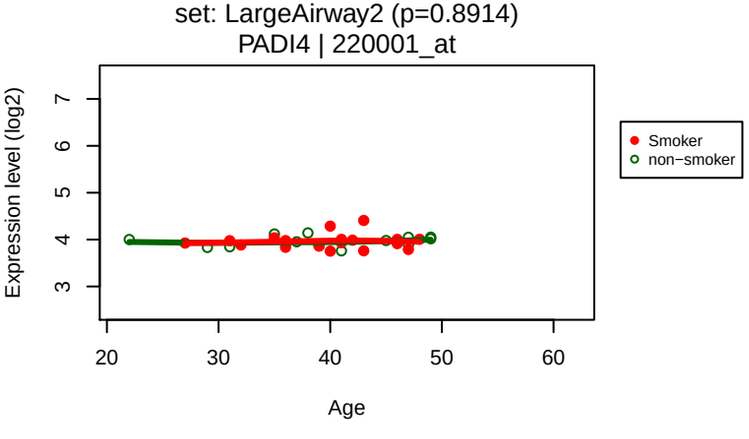

Gene expression ratio (smokers/nonsmokers)
gene: PADI4 | 220001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3254
p=0.3165
p=0.6175
p=0.3795
p=0.8914
p=0.5756
p=0.4512
p=0.5215
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

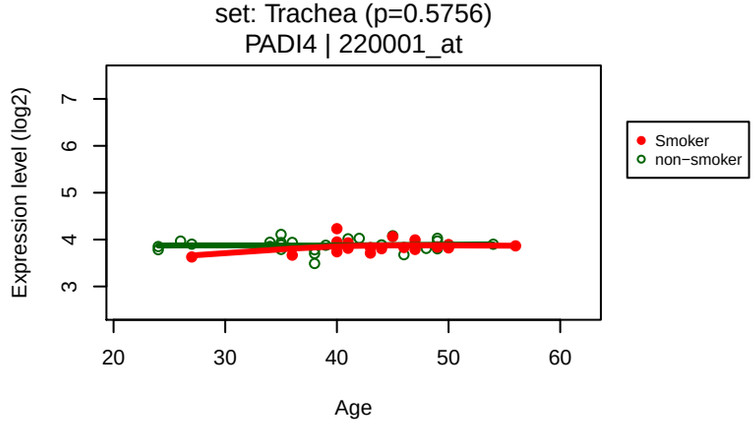

Gene expression ratio in Males
gene: PADI4 | 220001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2424
p=0.789
p=0.8049
p=0.2005
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PADI4 | 220001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8144
p=0.9369
p=0.9867
p=0.7181
SmallAirway2
LargeAirway2
Trachea
Macrophage2

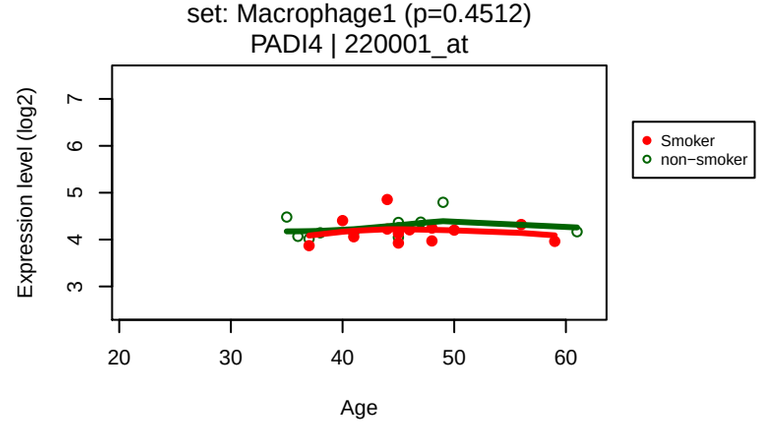

set: SmallAirway1 (p=0.3254)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker

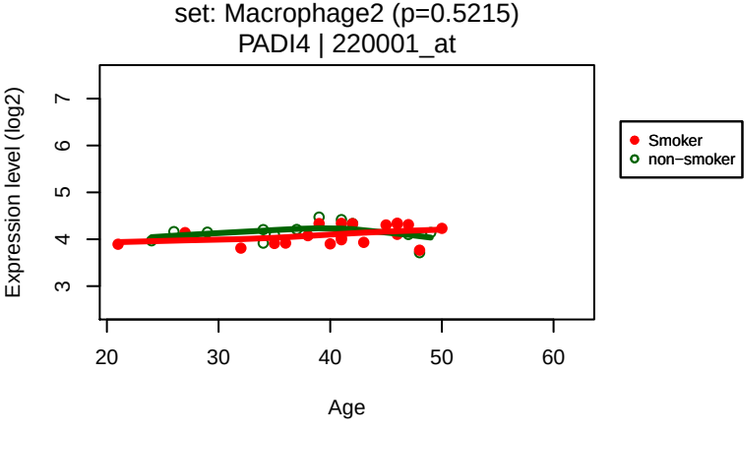

set: SmallAirway2 (p=0.3165)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker

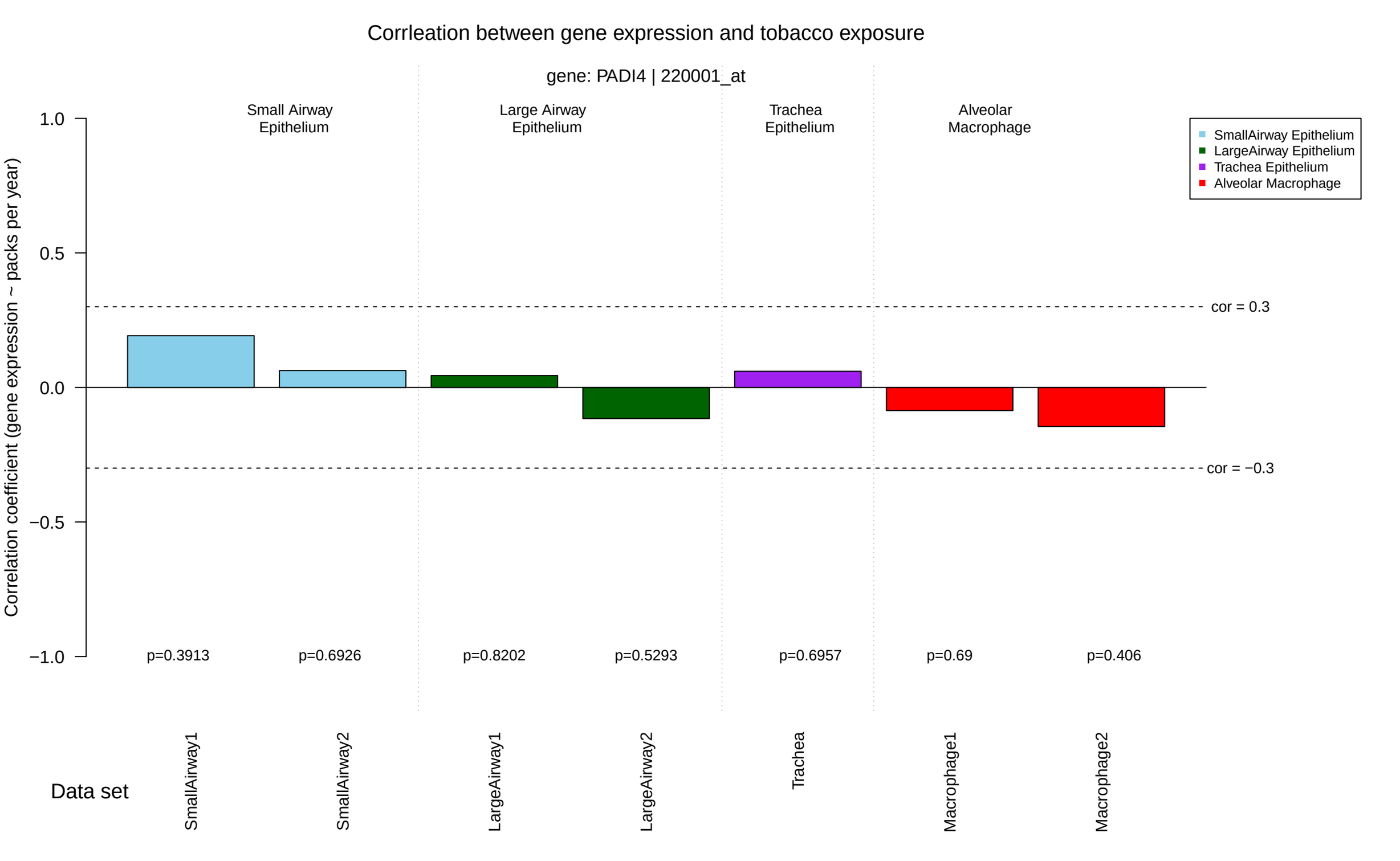

set: LargeAirway1 (p=0.3795)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8914)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5756)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4512)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5215)
PADI4 | 220001_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PADI4 | 220001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3913
p=0.6926
p=0.8202
p=0.5293
p=0.6957
p=0.69
p=0.406
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2